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Department Of Software Engineering

Cloud Computing Lab

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Section: 5-B

LAB NO 13

Task 0 Lab Setup (Codespace & GH CLI)

- task0_codespace_create_and_list.png

```
PS C:\Users\saira> gh repo create Lab13 --public
✓ Created repository 23-22411-057-sudo/Lab13 on github.com
https://github.com/23-22411-057-sudo/Lab13
PS C:\Users\saira> git clone https://github.com/23-22411-057-sudo/Lab13.git
Cloning into 'Lab13'...
warning: You appear to have cloned an empty repository.
PS C:\Users\saira> cd Lab13
PS C:\Users\saira\Lab13> echo "# Lab 13" > README.md
PS C:\Users\saira\Lab13> git add README.md
PS C:\Users\saira\Lab13> git commit -m "Initial commit with README"
[main (root-commit) 49365ad] Initial commit with README
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 README.md
PS C:\Users\saira\Lab13> git branch -M main
PS C:\Users\saira\Lab13> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 258 bytes | 86.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/23-22411-057-sudo/Lab13.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\saira\Lab13> gh codespace create --repo 23-22411-057-sudo/Lab13 --branch main
✓ Codespaces usage for this repository is paid for by 23-22411-057-sudo
? Choose Machine Type: 2 cores, 8 GB RAM, 32 GB storage
legendary-spoon-x5g59v9vj7v62vx6q
```

- task0_codespace_ssh_connected.png

```
PS C:\Users\saira\Lab13> gh codespace list
NAME              DISPLAY_NAME      REPOSITORY              BRANCH  STATE      CREATED AT
studious-fishstick-r454rprpjx94f6p7  studious fishstick  23-22411-057-sudo/exam_prep  main*   Shutdown   about 1 month ago
vigilant-system-5g7g4x4xw74737p4w    vigilant system     23-22411-057-sudo/CC_Saira_2023-BSE-057-Lab11  main*   Shutdown   about 3 days ago
vigilant-space-spork-q7579p9pwjq6cxj4j  vigilant space spork  23-22411-057-sudo/CC_Saira_2023-BSE-057-Lab12  main    Shutdown   about 2 days ago
legendary-spoon-x5g59v9vj7v62vx6q      legendary spoon     23-22411-057-sudo/Lab13      main    Available  less than a minute ago
PS C:\Users\saira\Lab13> gh codespace ssh -c legendary-spoon-x5g59v9vj7v62vx6q
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

@23-22411-057-sudo → /workspaces/Lab13 (main) $
```

Task 1 — Create IAM Group and Output Details

- task1_project_directory.png

```
@23-22411-057-sudo → /workspaces/Lab13 (main) $ mkdir -p ~/Lab13
@23-22411-057-sudo → /workspaces/Lab13 (main) $ cd ~/Lab13
@23-22411-057-sudo → ~/Lab13 $ |
```

- task1_file_created.png

```
@23-22411-057-sudo → ~/Lab13 $ touch main.tf
@23-22411-057-sudo → ~/Lab13 $ nano main.tf
@23-22411-057-sudo → ~/Lab13 $
```

- task1_main_tf.png

```
GNU nano 7.2 main.tf *
provider "aws" {
  shared_config_files      = ["~/aws/config"]
  shared_credentials_files = ["~/aws/credentials"]
}

resource "aws_iam_group" "developers" {
  name = "developers"
  path = "/groups/"
}

output "group_details" {
  value = {
    group_name = aws_iam_group.developers.name
    group_arn  = aws_iam_group.developers.arn
    unique_id  = aws_iam_group.developers.unique_id
  }
}
```

- task1_terraform_init.png

```
@23-22411-057-sudo → ~/Lab13 $ terraform init
```

Initializing the backend...

Initializing provider plugins...

- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.27.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

- task1_terraform_apply.png

```
@23-22411-057-sudo → ~/Lab13 $ terraform apply -auto-approve
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# aws_iam_group.developers will be created
+ resource "aws_iam_group" "developers" {
  + arn      = (known after apply)
  + id       = (known after apply)
  + name     = "developers"
  + path     = "/groups/"
  + unique_id = (known after apply)
}
```

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```
+ group_details = {
  + group_arn = (known after apply)
  + group_name = "developers"
  + unique_id = (known after apply)
}
```

aws_iam_group.developers: Creating...

aws_iam_group.developers: Creation complete after 1s [id=developers]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

```
group_details = {
  "group_arn" = "arn:aws:iam::359416636630:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAVHLXGUTLP6YA05TDZ"
}
```

- task1_terraform_output.png

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

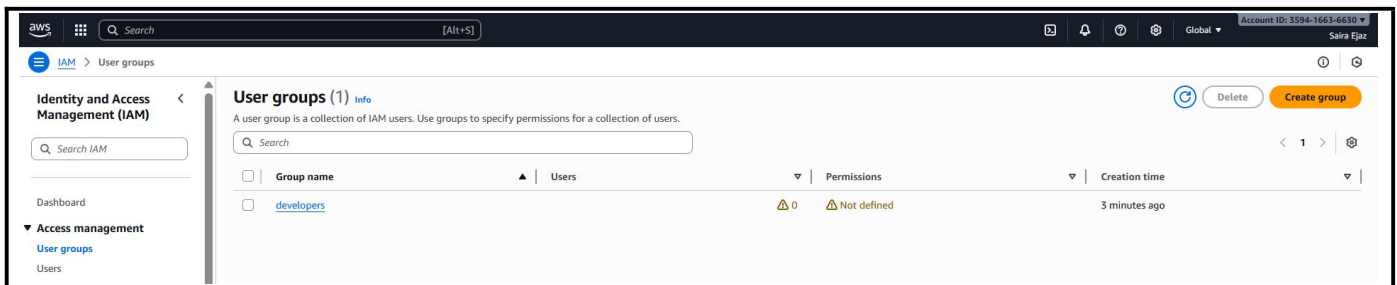
```
group_details = {
  "group_arn" = "arn:aws:iam::359416636630:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAVHLXGUTLP6YA05TDZ"
}
```

```
@23-22411-057-sudo → ~/Lab13 $ terraform output
```

```
group_details = {
  "group_arn" = "arn:aws:iam::359416636630:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAVHLXGUTLP6YA05TDZ"
}
```

```
@23-22411-057-sudo → ~/Lab13 $
```

- task1_aws_console_group.png



Task 2 — Create IAM User with Group Membership

- task2_main_tf_user.png

```
GNU nano 7.2 main.tf *
provider "aws" {
  shared_config_files = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

# IAM Group
resource "aws_iam_group" "developers" {
  name = "developers"
  path = "/groups/"
}

output "group_details" {
  value = {
    group_name = aws_iam_group.developers.name
    group_arn = aws_iam_group.developers.arn
    unique_id = aws_iam_group.developers.unique_id
  }
}

# IAM User
resource "aws_iam_user" "lb" {
  name = "loadbalancer"
  path = "/users/"
  force_destroy = true
  tags = {
    DisplayName = "Load Balancer"
  }
}

# Add User to Group
resource "aws_iam_user_group_membership" "lb_membership" {
  user = aws_iam_user.lb.name
  groups = [
    aws_iam_group.developers.name
  ]
}

# Output User Details
output "user_details" {
  value = {
    user_name = aws_iam_user.lb.name
  }
}
```

- task2_terraform_apply.png

```
@23-22411-057-sudo → ~/Lab13 $ terraform apply -auto-approve
aws_iam_group.developers: Refreshing state... [id=developers]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_iam_user.lb will be created
+ resource "aws_iam_user" "lb" {
  + arn           = (known after apply)
  + force_destroy = true
  + id            = (known after apply)
  + name          = "loadbalancer"
  + path          = "/users/"
  + tags          = {
    + "DisplayName" = "Load Balancer"
  }
  + tags_all      = {
    + "DisplayName" = "Load Balancer"
  }
  + unique_id     = (known after apply)
}

# aws_iam_user_group_membership.lb_membership will be created
+ resource "aws_iam_user_group_membership" "lb_membership" {
  + groups = [
    + "developers",
  ]
  + id      = (known after apply)
  + user    = "loadbalancer"
}

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ user_details = {
  + unique_id = (known after apply)
  + user_arn  = (known after apply)
  + user_name = "loadbalancer"
}

aws_iam_user.lb: Creating...
aws_iam_user.lb: Creation complete after 1s [id=loadbalancer]
aws_iam_user_group_membership.lb_membership: Creating...
aws_iam_user_group_membership.lb_membership: Creation complete after 1s [id=terraform-202601051804336217000000001]
```

- task2_terraform_output.png

```
Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

group_details = {
  "group_arn" = "arn:aws:iam::359416636630:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAVHLXGUTLP6YAO5TDZ"
}
user_details = {
  "unique_id" = "AIDAVHLXGUTLCWOZHPKOB"
  "user_arn" = "arn:aws:iam::359416636630:user/users/loadbalancer"
  "user_name" = "loadbalancer"
}
@23-22411-057-sudo → ~/Lab13 $ terraform output

group_details = {
  "group_arn" = "arn:aws:iam::359416636630:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAVHLXGUTLP6YAO5TDZ"
}
user_details = {
  "unique_id" = "AIDAVHLXGUTLCWOZHPKOB"
  "user_arn" = "arn:aws:iam::359416636630:user/users/loadbalancer"
  "user_name" = "loadbalancer"
}
@23-22411-057-sudo → ~/Lab13 $
@23-22411-057-sudo → ~/Lab13 $
```

- task2_aws_console_user.png

Users (4) Info

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Q Search

< 1 >

☐	User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID	Active key age	Acce
☐	Admin	/	0	5 days ago	-	38 days	38 days ago	Active - AKIAVHLXGUT...	10 days	5
☐	loadbalancer	/users/	1	-	-	-	-	-	-	-
☐	User2	/	0	3 days ago	-	-	-	Active - AKIAVHLXGUT...	3 days	3
☐	User3	/	0	13 minutes ago	-	-	-	Active - AKIAVHLXGUT...	2 days	1

- task2_aws_console_user_groups.png

loadbalancer Info

Delete

Summary

ARN

arn:aws:iam::359416636630:user/users/loadbalancer

Console access

Disabled

Access key 1

Create access key

Created

January 05, 2026, 23:04 (UTC+05:00)

Last console sign-in

-

Permissions

Groups (1)

Tags (1)

Security credentials

Last Accessed

User groups membership

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users. A user can be a member of up to 10 groups at a time.

Remove

Add user to groups

☐

Group name

Attached policies 2

☐

developers

-

Identity and Access Management (IAM)

Q Search IAM

Dashboard

Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management

Temporary delegation requests

Access reports

Access Analyzer

Resource analysis

Unused access

Analyzer settings

Credential report

User groups (1) Info

Delete

Create group

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

Q Search

< 1 >

☐	Group name	Users	Permissions	Creation time
☐	developers	1	Not defined	13 minutes ago

Task 3 — Attach Policies to IAM Group

- task3_main_tf_policies.png

```

GNU nano 7.2                                     main.tf *
}
}

# IAM User
resource "aws_iam_user" "lb" {
  name = "loadbalancer"
  path = "/users/"
  force_destroy = true
  tags = {
    DisplayName = "Load Balancer"
  }
}

# Add User to Group
resource "aws_iam_user_group_membership" "lb_membership" {
  user = aws_iam_user.lb.name
  groups = [
    aws_iam_group.developers.name
  ]
}

# Output User Details
output "user_details" {
  value = {
    user_name = aws_iam_user.lb.name
    user_arn = aws_iam_user.lb.arn
    unique_id = aws_iam_user.lb.unique_id
  }
}

# Attach AmazonEC2FullAccess to developers group
resource "aws_iam_group_policy_attachment" "developer_ec2_fullaccess" {
  group = aws_iam_group.developers.name
  policy_arn = "arn:aws:iam::aws:policy/AmazonEC2FullAccess"
}

# Attach IAMUserChangePassword to developers group
resource "aws_iam_group_policy_attachment" "change_password" {
  group = aws_iam_group.developers.name
  policy_arn = "arn:aws:iam::aws:policy/IAMUserChangePassword"
}

```

- task3_terraform_apply.png

```

@23-22411-057-sudo → ~/Lab13 $ terraform apply -auto-approve
aws_iam_group.developers: Refreshing state... [id=developers]
aws_iam_user.lb: Refreshing state... [id=loadbalancer]
aws_iam_user_group_membership.lb_membership: Refreshing state... [id=terraform-20260105180433621700000001]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_iam_group_policy_attachment.change_password will be created
+ resource "aws_iam_group_policy_attachment" "change_password" {
  + group = "developers"
  + id = (known after apply)
  + policy_arn = "arn:aws:iam::aws:policy/IAMUserChangePassword"
}

# aws_iam_group_policy_attachment.developer_ec2_fullaccess will be created
+ resource "aws_iam_group_policy_attachment" "developer_ec2_fullaccess" {
  + group = "developers"
  + id = (known after apply)
  + policy_arn = "arn:aws:iam::aws:policy/AmazonEC2FullAccess"
}

Plan: 2 to add, 0 to change, 0 to destroy.
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Creating...
aws_iam_group_policy_attachment.change_password: Creating...
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Creation complete after 1s [id=developers-20260105181032553400000002]
aws_iam_group_policy_attachment.change_password: Creation complete after 1s [id=developers-20260105181032548500000001]

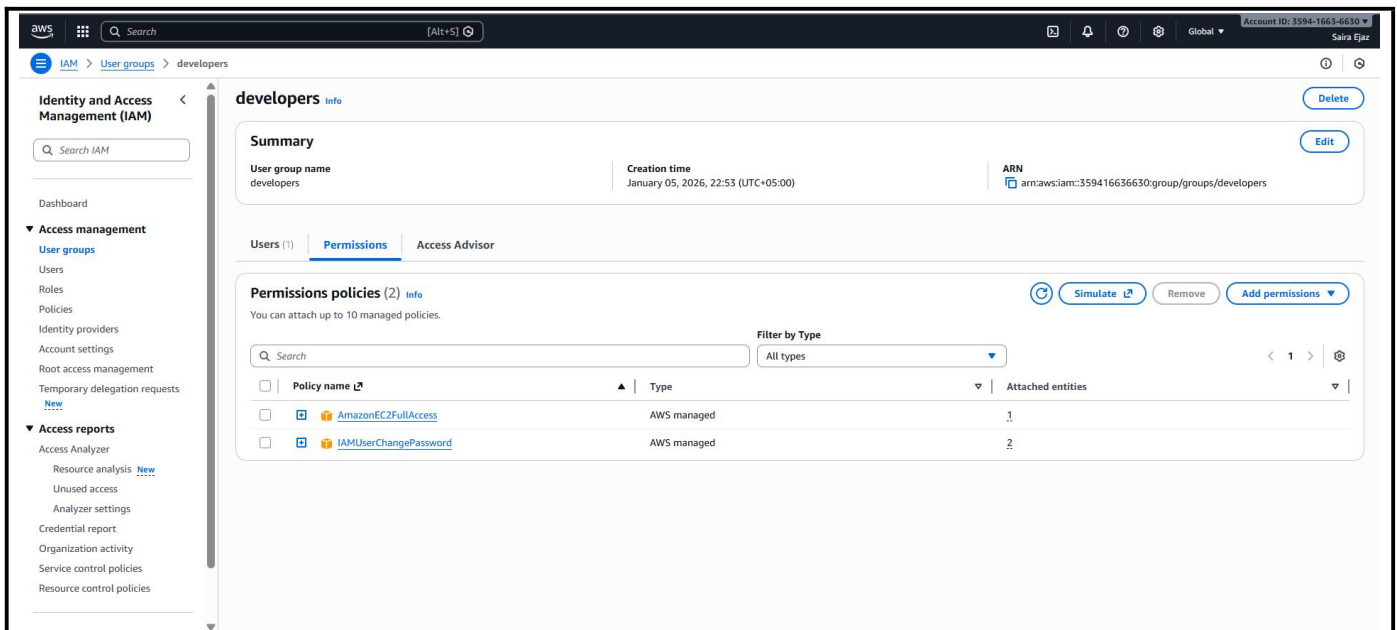
Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

group_details = {
  "group_arn" = "arn:aws:iam::359416636630:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAVHLXGUTLP6YA05TDZ"
}
user_details = {
  "unique_id" = "AIDAVHLXGUTLCWOZHPKOB"
  "user_arn" = "arn:aws:iam::359416636630:user/users/loadbalancer"
  "user_name" = "loadbalancer"
}
@23-22411-057-sudo → ~/Lab13 $

```

- task3_aws_console_policies.png



The screenshot shows the AWS IAM console for the 'developers' user group. The 'Permissions' tab is selected, showing two attached policies: 'AmazonEC2FullAccess' and 'IAMUserChangePassword'. The left sidebar shows the navigation menu with 'Access management' expanded.

Task 4 — Create Login Profile for IAM User

- task4_variables_tf.png

```
GNU nano 7.2 variables.tf *
variable "iam_password" {
  description = "Temporary password for the IAM user"
  type        = string
  sensitive   = true
  default     = "1dontKnow"
}
```

- task4_create_login_script.png

```
GNU nano 7.2 create-login-profile.sh *
#!/usr/bin/env bash
set -euo pipefail

USERNAME="$1"
PASSWORD="$2"

# Check if login profile already exists
if aws iam get-login-profile --user-name "$USERNAME" >/dev/null 2>&1; then
  echo "Login profile already exists for $USERNAME. Skipping."
else
  echo "Creating login profile for $USERNAME"
  aws iam create-login-profile \
    --user-name "$USERNAME" \
    --password "$PASSWORD" \
    --password-reset-required
fi
```

- task4_chmod_script.png

```
@23-22411-057-sudo → ~/Lab13 $ chmod +x create-login-profile.sh
@23-22411-057-sudo → ~/Lab13 $ |
```

- task4_main_tf_login_profile.png

```

GNU nano 7.2                                     main.tf *
provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

# IAM Group
resource "aws_iam_group" "developers" {
  name = "developers"
  path = "/groups/"
}

output "group_details" {
  value = {
    group_name = aws_iam_group.developers.name
    group_arn  = aws_iam_group.developers.arn
    unique_id  = aws_iam_group.developers.unique_id
  }
}

# IAM User
resource "aws_iam_user" "lb" {
  name = "loadbalancer"
  path = "/users/"
  force_destroy = true
  tags = {
    DisplayName = "Load Balancer"
  }
}

resource "null_resource" "create_login_profile" {
  triggers = {
    password_hash = sha256(var.iam_password)
    user          = aws_iam_user.lb.name
  }

  depends_on = [aws_iam_user.lb]

  provisioner "local-exec" {
    command = "${path.module}/create-login-profile.sh ${aws_iam_user.lb.name} '${var.iam_password}'"
  }
}

```

● task4_terraform_apply.png

```

@23-22411-057-sudo → ~/Lab13 $ terraform apply -auto-approve -var="iam_password=MySecurePass123!"
aws_iam_group.developers: Refreshing state... [id=developers]
aws_iam_user.lb: Refreshing state... [id=loadbalancer]
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Refreshing state... [id=developers-20260105181032553400000002]
aws_iam_group_policy_attachment.change_password: Refreshing state... [id=developers-20260105181032548500000001]
aws_iam_user_group_membership.lb_membership: Refreshing state... [id=terraform-20260105180433621700000001]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# null_resource.create_login_profile will be created
+ resource "null_resource" "create_login_profile" {
  + id          = (known after apply)
  + triggers = {
    + "password_hash" = (sensitive value)
    + "user"          = "loadbalancer"
  }
}

Plan: 1 to add, 0 to change, 0 to destroy.
null_resource.create_login_profile: Creating...
null_resource.create_login_profile: Provisioning with 'local-exec'...
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile: Creation complete after 3s [id=8089229781452210809]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:
group_details = {
  "group_arn" = "arn:aws:iam::359416636630:group/groups/developers"
}

```

● task4_aws_cli_verify.png

```

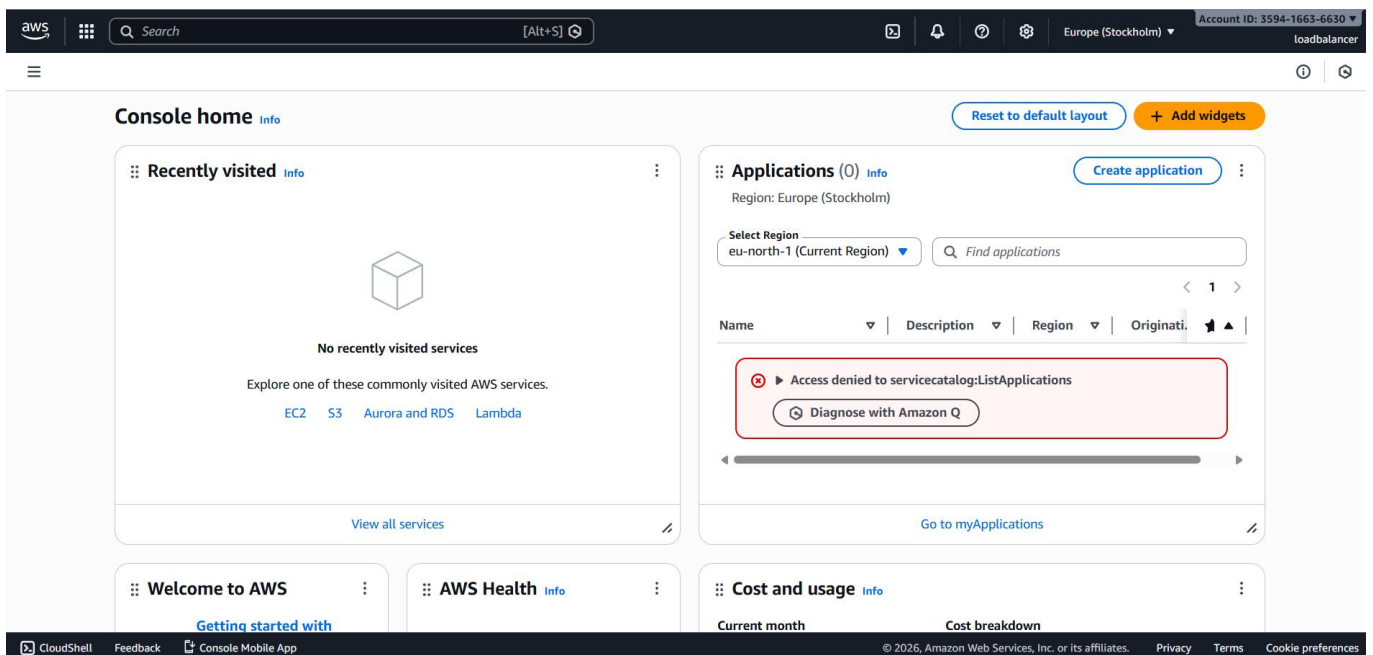
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

group_details = {
  "group_arn" = "arn:aws:iam::359416636630:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAVHLXGUTLP6YA05TDZ"
}
user_details = {
  "unique_id" = "AIDAVHLXGUTLCWOZHPK0B"
  "user_arn" = "arn:aws:iam::359416636630:user/users/loadbalancer"
  "user_name" = "loadbalancer"
}
@23-22411-057-sudo → ~/Lab13 $ aws iam get-login-profile --user-name loadbalancer
{
  "LoginProfile": {
    "UserName": "loadbalancer",
    "CreateDate": "2026-01-05T18:16:56+00:00",
    "PasswordResetRequired": true
  }
}
@23-22411-057-sudo → ~/Lab13 $

```

- task4_aws_console_login.png



- task4_aws_console_password_reset.png



Password reset ⓘ

Your account (**359416636630**) password has expired or requires a reset.

To continue, please verify your old and set a new password for **loadbalancer** (not you?).

Old Password

.....

☐ Show Password

New Password

Confirm New Password

☐ Show Password

Confirm Password Change

[Sign in to a different account](#)



✔ Password reset successful

Password reset

Your IAM User (**loadbalancer**) password has been reset successfully.

Continue to sign in

Task 5 — Generate Access Keys for IAM User

- task5_main_tf_access_keys.png

```
GNU nano 7.2 main.tf *
user_arn = aws_iam_user.lb.arn
unique_id = aws_iam_user.lb.unique_id
}
}

# =====
# Task 4: Login Profile
# =====
resource "null_resource" "create_login_profile" {
  triggers = {
    password_hash = sha256(var.iam_password)
    user          = aws_iam_user.lb.name
  }

  depends_on = [aws_iam_user.lb]

  provisioner "local-exec" {
    command = "${path.module}/create-login-profile.sh ${aws_iam_user.lb.name} '${var.iam_password}'"
  }
}

# =====
# Task 5: IAM Access Keys
# =====
resource "aws_iam_access_key" "lb_access_key" {
  user = aws_iam_user.lb.name
}

output "access_key_id" {
  value = aws_iam_access_key.lb_access_key.id
}

output "access_key_secret" {
  value     = aws_iam_access_key.lb_access_key.secret
  sensitive = true
}
```

- task5_terraform_apply.png

```
@23-22411-057-sudo → ~/Lab13 $ terraform apply -auto-approve -var="iam_password=MySecurePass123!"
aws_iam_group.developers: Refreshing state... [id=developers]
aws_iam_user.lb: Refreshing state... [id=loadbalancer]
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Refreshing state... [id=developers-20260105181032553400000002]
aws_iam_group_policy_attachment.change_password: Refreshing state... [id=developers-20260105181032548500000001]
aws_iam_user_group_membership.lb_membership: Refreshing state... [id=terraform-20260105180433621700000001]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# null_resource.create_login_profile will be created
+ resource "null_resource" "create_login_profile" {
  + id          = (known after apply)
  + triggers = {
    + "password_hash" = (sensitive value)
    + "user"          = "loadbalancer"
  }
}

Plan: 1 to add, 0 to change, 0 to destroy.
null_resource.create_login_profile: Creating...
null_resource.create_login_profile: Provisioning with 'local-exec'...
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile: Creation complete after 3s [id=8089229781452210809]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:
group_details = {
  "group_arn" = "arn:aws:iam::359416636630:group/groups/developers"
```

- task5_terraform_output.png

Outputs:

```
access_key_id = "AKIAVHLXGUTLNNKEBDUT"
access_key_secret = <sensitive>
group_details = {
  "group_arn" = "arn:aws:iam::359416636630:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAVHLXGUTLP6YA05TDZ"
}
user_details = {
  "unique_id" = "AIDAVHLXGUTLCWOZHPKOB"
  "user_arn" = "arn:aws:iam::359416636630:user/users/loadbalancer"
  "user_name" = "loadbalancer"
}
@23-22411-057-sudo → ~/Lab13 $ terraform output
access_key_id = "AKIAVHLXGUTLNNKEBDUT"
access_key_secret = <sensitive>
group_details = {
  "group_arn" = "arn:aws:iam::359416636630:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAVHLXGUTLP6YA05TDZ"
}
user_details = {
  "unique_id" = "AIDAVHLXGUTLCWOZHPKOB"
  "user_arn" = "arn:aws:iam::359416636630:user/users/loadbalancer"
  "user_name" = "loadbalancer"
}
@23-22411-057-sudo → ~/Lab13 $ |
```

- task5_tfstate_secret. Png

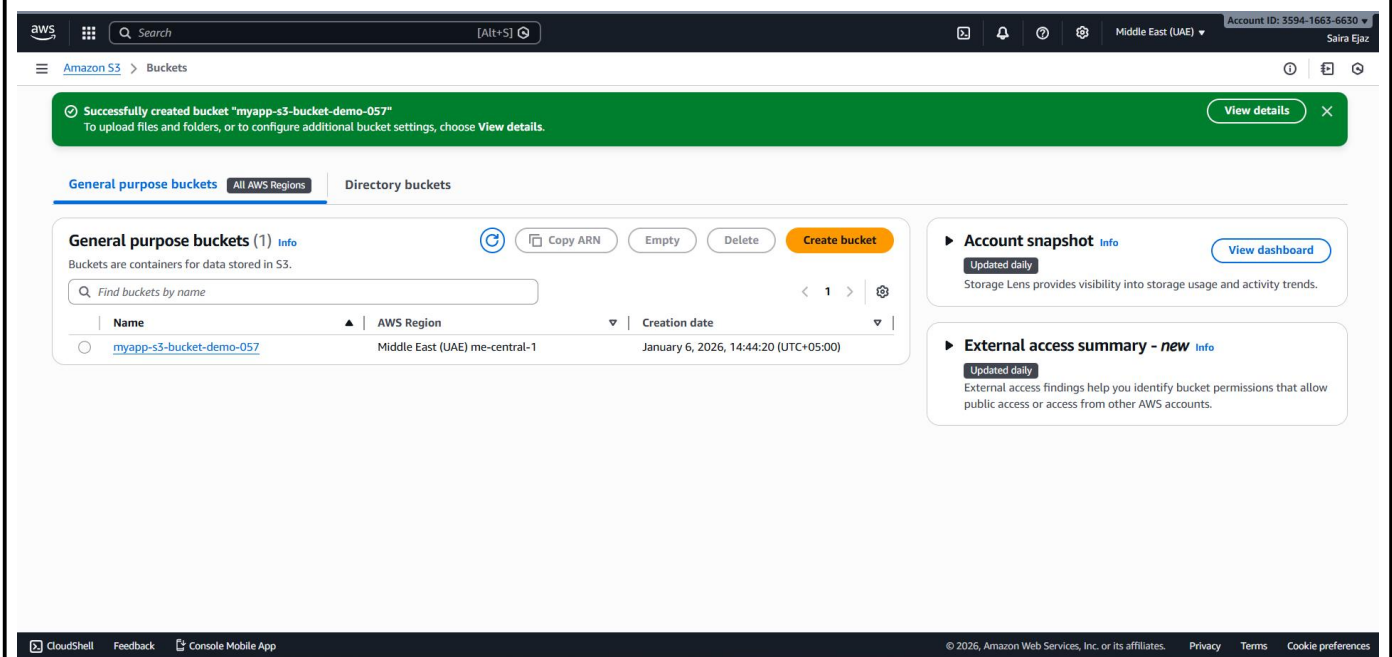
```
@23-22411-057-sudo → ~/Lab13 $ cat terraform.tfstate | grep -A 10 "access_key_secret"
  "access_key_secret": {
    "value": "yM3qnSg2jpA6knnQMxB22SQhXMmThGW88C2+EEe8",
    "type": "string",
    "sensitive": true
  },
  "group_details": {
    "value": {
      "group_arn": "arn:aws:iam::359416636630:group/groups/developers",
      "group_name": "developers",
      "unique_id": "AGPAVHLXGUTLP6YA05TDZ"
    },
  },
@23-22411-057-sudo → ~/Lab13 $ |
```

- task5_aws_console_access_keys. Png

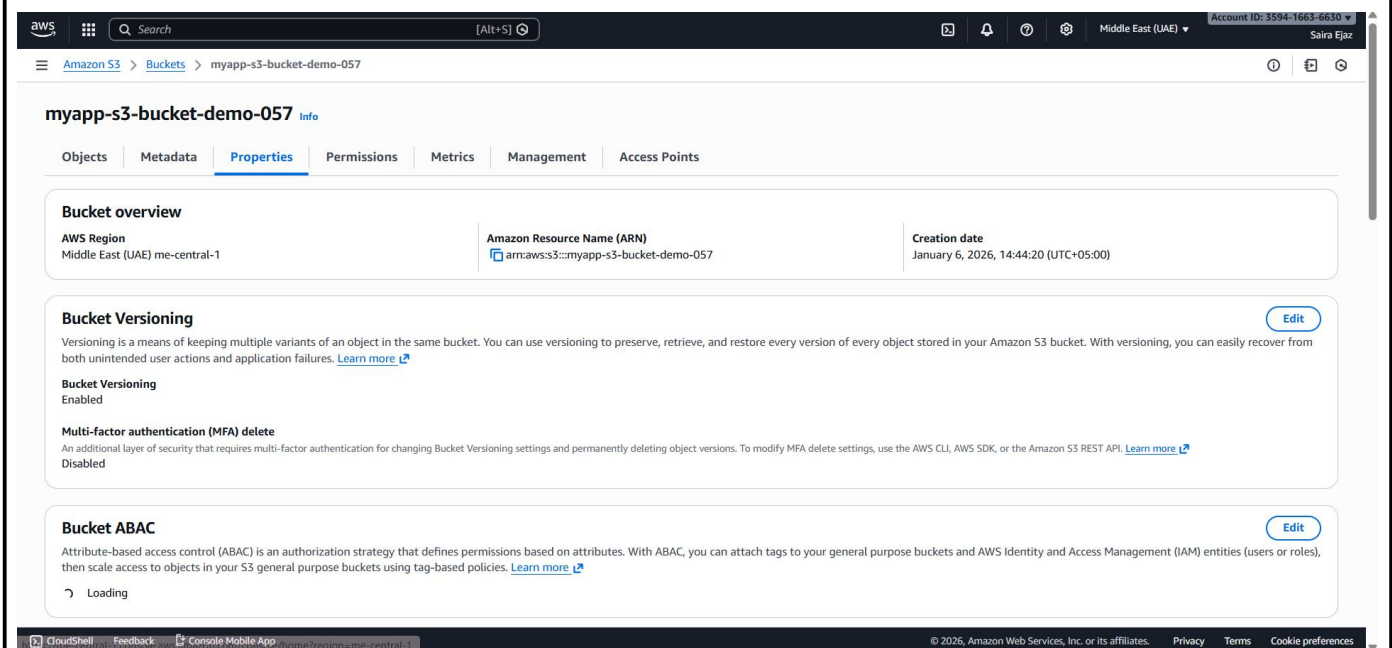
The screenshot shows the AWS IAM console interface for a user named 'loadbalancer'. The left sidebar contains navigation links for Identity and Access Management (IAM), Access management, and Access reports. The main content area displays the user's summary, including their ARN, console access status, and created date. Below the summary, there are tabs for Permissions, Groups, Tags, Security credentials, and Last Accessed. The Permissions tab is active, showing a table of permissions policies attached to the user. The table lists two policies: 'AmazonEC2FullAccess' and 'IAMUserChangePassword', both managed by AWS and attached to the 'developers' group. At the bottom, there is a section for 'Permissions boundary' and a link to 'Generate policy based on CloudTrail events'.

Task 6 — Implement Terraform Remote State with S3

● task6_s3_bucket_create.png



● task6_s3_bucket_versioning.png



● task6_main_tf_backend.png

```

GNU nano 7.2                                     main.tf *
terraform {
  backend "s3" {
    bucket     = "myapp-s3-bucket-demo-057"
    key        = "myapp/terraform.tfstate"
    region     = "me-central-1"
    encrypt    = true
    use_lockfile = true
  }
}
provider "aws" {
  shared_config_files = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

# =====
# Task 1 & 3: IAM Group
# =====
resource "aws_iam_group" "developers" {
  name = "developers"
  path = "/groups/"
}

# Attach AWS Managed Policies (Task 3)
resource "aws_iam_group_policy_attachment" "developer_ec2_fullaccess" {
  group       = aws_iam_group.developers.name
  policy_arn = "arn:aws:iam::aws:policy/AmazonEC2FullAccess"
}

resource "aws_iam_group_policy_attachment" "change_password" {
  group       = aws_iam_group.developers.name
  policy_arn = "arn:aws:iam::aws:policy/IAMUserChangePassword"
}

output "group_details" {
  value = {
    group_name = aws_iam_group.developers.name
    group_arn  = aws_iam_group.developers.arn
  }
}

```

- task6_terraform_init_migrate.png

```

@23-22411-057-sudo → ~/Lab13 $ terraform init -migrate-state

Initializing the backend...
Do you want to copy existing state to the new backend?
Pre-existing state was found while migrating the previous "local" backend to the
newly configured "s3" backend. No existing state was found in the newly
configured "s3" backend. Do you want to copy this state to the new "s3"
backend? Enter "yes" to copy and "no" to start with an empty state.

Enter a value: yes

Successfully configured the backend "s3"! Terraform will automatically
use this backend unless the backend configuration changes.

Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/null from the dependency lock file
- Using previously-installed hashicorp/aws v6.27.0
- Using previously-installed hashicorp/null v3.2.4

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@23-22411-057-sudo → ~/Lab13 $ |

```

- task6_terraform_apply.png

```
@23-22411-057-sudo → ~/Lab13 $ terraform apply -auto-approve -var="iam_password=MySecurePass123!"
aws_iam_group.developers: Refreshing state... [id=developers]
aws_iam_user.lb: Refreshing state... [id=loadbalancer]
aws_iam_group_policy_attachment.change_password: Refreshing state... [id=developers-20260105181032548500000001]
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Refreshing state... [id=developers-20260105181032553400000002]
null_resource.create_login_profile: Refreshing state... [id=2562651954893160502]
aws_iam_access_key.lb_access_key: Refreshing state... [id=AKIAVHLXGUTLNNKEBDUT]
aws_iam_user_group_membership.lb_membership: Refreshing state... [id=terraform-20260105180433621700000001]
```

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```
access_key_id = "AKIAVHLXGUTLNNKEBDUT"
access_key_secret = <sensitive>
group_details = {
  "group_arn" = "arn:aws:iam::359416636630:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAVHLXGUTLP6YA05TDZ"
}
user_details = {
  "unique_id" = "AIDAVHLXGUTLCWOZHPKOB"
  "user_arn" = "arn:aws:iam::359416636630:user/users/loadbalancer"
  "user_name" = "loadbalancer"
}
```

```
@23-22411-057-sudo → ~/Lab13 $ |
```

- task6_s3_tfstate_file.png

The screenshot shows the AWS Management Console interface for an Amazon S3 bucket named 'myapp-s3-bucket-demo-057'. The left sidebar shows the 'Amazon S3' navigation menu with options like Buckets, Access management and security, and Storage management and insights. The main content area displays the 'myapp/' prefix with a 'Copy S3 URI' button. Below this, the 'Objects' tab is active, showing a table of objects. There is one object named 'terraform.tfstate' with a size of 6.2 KB and a storage class of 'Standard'. The table headers include Name, Type, Last modified, Size, and Storage class.

- task6_local_state_backup.png

```
@23-22411-057-sudo → ~/Lab13 $ ls -la terraform.tfstate*
-rw-rw-r-- 1 codespace codespace 0 Jan 6 09:50 terraform.tfstate
-rw-rw-r-- 1 codespace codespace 6315 Jan 6 09:50 terraform.tfstate.backup
@23-22411-057-sudo → ~/Lab13 $
```

- task6_terraform_destroy.png

```
@23-22411-057-sudo → ~/Lab13 $ terraform destroy -auto-approve
aws_iam_group.developers: Refreshing state... [id=developers]
aws_iam_user.lb: Refreshing state... [id=loadbalancer]
null_resource.create_login_profile: Refreshing state... [id=2562651954893160502]
aws_iam_access_key.lb_access_key: Refreshing state... [id=AKIAVHLXGUTLNNKEBDUT]
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Refreshing state... [id=developers-20260105181032553400000002]
aws_iam_group_policy_attachment.change_password: Refreshing state... [id=developers-20260105181032548500000001]
aws_iam_user_group_membership.lb_membership: Refreshing state... [id=terraform-20260105180433621700000001]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy
```

Terraform will perform the following actions:

```
# aws_iam_access_key.lb_access_key will be destroyed
- resource "aws_iam_access_key" "lb_access_key" {
  - create_date      = "2026-01-05T18:34:33Z" -> null
  - id               = "AKIAVHLXGUTLNNKEBDUT" -> null
  - secret           = (sensitive value) -> null
  - ses_smtp_password_v4 = (sensitive value) -> null
  - status           = "Active" -> null
  - user             = "loadbalancer" -> null
}

# aws_iam_group.developers will be destroyed
- resource "aws_iam_group" "developers" {
  - arn          = "arn:aws:iam::359416636630:group/groups/developers" -> null
  - id           = "developers" -> null
  - name         = "developers" -> null
  - path         = "/groups/" -> null
  - unique_id    = "AGPAVHLXGUTLP6YAOSTDZ" -> null
}

# aws_iam_group_policy_attachment.change_password will be destroyed
- resource "aws_iam_group_policy_attachment" "change_password" {
  - group      = "developers" -> null
  - id         = "developers-20260105181032548500000001" -> null
  - policy_arn = "arn:aws:iam::aws:policy/IAMUserChangePassword" -> null
}

# aws_iam_group_policy_attachment.developer_ec2_fullaccess will be destroyed
- resource "aws_iam_group_policy_attachment" "developer_ec2_fullaccess" {
  - group      = "developers" -> null
  - id         = "developers-20260105181032553400000002" -> null
  - policy_arn = "arn:aws:iam::aws:policy/AmazonEC2FullAccess" -> null
}

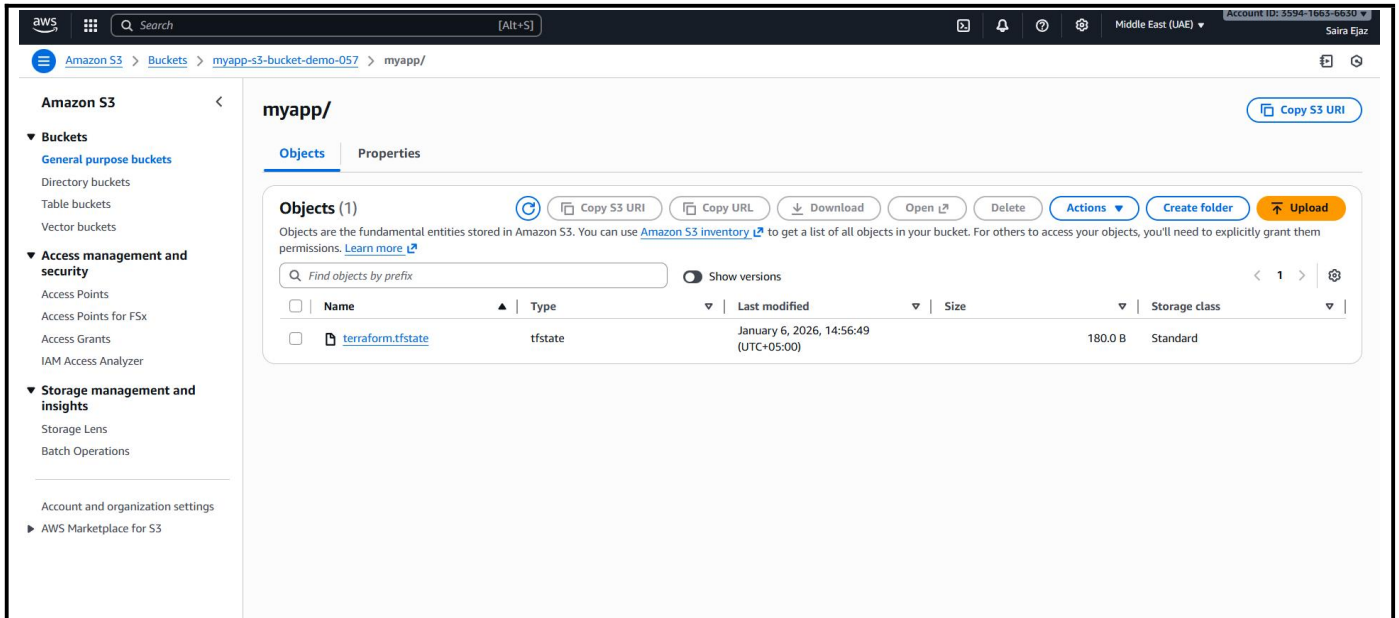
# aws_iam_user.lb will be destroyed
- resource "aws_iam_user" "lb" {
  - arn          = "arn:aws:iam::359416636630:user/users/loadbalancer" -> null
  - force_destroy = true -> null
  - id           = "loadbalancer" -> null
  - name         = "loadbalancer" -> null
  - path         = "/users/" -> null
  - tags         = {
    - "DisplayName" = "Load Balancer"
  }
```

- task6_s3_tfstate_destroyed. Png

```
j -> null
null_resource.create_login_profile: Destroying... [id=2562651954893160502]
null_resource.create_login_profile: Destruction complete after 0s
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Destroying... [id=developers-20260105181032553400000002]
aws_iam_group_policy_attachment.change_password: Destroying... [id=developers-20260105181032548500000001]
aws_iam_access_key.lb_access_key: Destroying... [id=AKIAVHLXGUTLNNKEBDUT]
aws_iam_user_group_membership.lb_membership: Destroying... [id=terraform-20260105180433621700000001]
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Destruction complete after 1s
aws_iam_user_group_membership.lb_membership: Destruction complete after 1s
aws_iam_group_policy_attachment.change_password: Destruction complete after 1s
aws_iam_group.developers: Destroying... [id=developers]
aws_iam_access_key.lb_access_key: Destruction complete after 1s
aws_iam_user.lb: Destroying... [id=loadbalancer]
aws_iam_group.developers: Destruction complete after 0s
aws_iam_user.lb: Destruction complete after 3s

Destroy complete! Resources: 7 destroyed.
@23-22411-057-sudo → ~/Lab13 $
```

- task6_s3_tfstate_destroyed. png



Task 7 — Create Multiple Users from CSV File

- task7_locals_tf.png

```
GNU nano 7.2                                locals.tf *
locals {
  users = csvdecode(file("${path.module}/users.csv"))
}
```

- task7_users_csv.png

```
GNU nano 7.2                                users.csv *
user_name
Michael
Dwight
Jim
Pam
Ryan
Andy
Robert
Stanley
Kevin
Angela
Oscar
Phyllis
Toby
Kelly
Darryl
Creed
Meredith
Erin
Gabe
Jan
David
Holly
Charles
Jo
Clark
Peter
```

- task7_main_tf_multiple_users.png

```
GNU nano 7.2 main.tf *
#####
# Terraform Backend (S3 Remote State)
#####
terraform {
  backend "s3" {
    bucket = "myapp-s3-bucket-demo"
    key    = "myapp/terraform.tfstate"
    region = "me-central-1"
    encrypt = true
  }
}

#####
# AWS Provider
#####
provider "aws" {
  region = "me-central-1"
}

#####
# Variables
#####
variable "iam_password" {
  description = "Password for IAM users"
  type        = string
  sensitive   = true
}

#####
# IAM Group
#####
resource "aws_iam_group" "developers" {
  name = "developers"
  path = "/groups/"
}

#####
```

```
GNU nano 7.2 main.tf *

#####
# Create Multiple IAM Users from CSV
#####
resource "aws_iam_user" "users" {
  for_each = { for user in local.users : user.user_name => user }

  name      = each.value.user_name
  path      = "/users/"
  force_destroy = true

  tags = {
    DisplayName = each.value.user_name
    CreatedBy   = "Terraform"
  }
}

#####
# Add Users to Developers Group
#####
resource "aws_iam_user_group_membership" "users_membership" {
  for_each = aws_iam_user.users

  user = each.value.name
  groups = [
    aws_iam_group.developers.name
  ]
}

#####
# Create Access Keys for Users
#####
resource "aws_iam_access_key" "users_access_keys" {
  for_each = aws_iam_user.users
  user     = each.value.name
}
```

- task7_terraform_init.png

```
@23-22411-057-sudo → ~/Lab13 $ terraform init -reconfigure
```

Initializing the backend...

Successfully configured the backend "s3"! Terraform will automatically use this backend unless the backend configuration changes.

Initializing provider plugins...

- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.27.0

Terraform has made some changes to the provider dependency selections recorded in the .terraform.lock.hcl file. Review those changes and commit them to your version control system if they represent changes you intended to make.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
@23-22411-057-sudo → ~/Lab13 $
```

- task7_terraform_apply.png

```
@23-22411-057-sudo → ~/Lab13 $ terraform apply -auto-approve -var="iam_password=MySecurePass123!"
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# aws_iam_access_key.users_access_keys["Andy"] will be created
+ resource "aws_iam_access_key" "users_access_keys" {
  + create_date           = (known after apply)
  + encrypted_secret       = (known after apply)
  + encrypted_ses_smtp_password_v4 = (known after apply)
  + id                    = (known after apply)
  + key_fingerprint       = (known after apply)
  + secret                = (sensitive value)
  + ses_smtp_password_v4  = (sensitive value)
  + status                = "Active"
  + user                  = "Andy"
}
```

```
# aws_iam_access_key.users_access_keys["Angela"] will be created
+ resource "aws_iam_access_key" "users_access_keys" {
  + create_date           = (known after apply)
  + encrypted_secret       = (known after apply)
  + encrypted_ses_smtp_password_v4 = (known after apply)
  + id                    = (known after apply)
  + key_fingerprint       = (known after apply)
  + secret                = (sensitive value)
  + ses_smtp_password_v4  = (sensitive value)
  + status                = "Active"
  + user                  = "Angela"
}
```

```
# aws_iam_access_key.users_access_keys["Charles"] will be created
+ resource "aws_iam_access_key" "users_access_keys" {
  + create_date           = (known after apply)
  + encrypted_secret       = (known after apply)
  + encrypted_ses_smtp_password_v4 = (known after apply)
  + id                    = (known after apply)
  + key_fingerprint       = (known after apply)
  + secret                = (sensitive value)
  + ses_smtp_password_v4  = (sensitive value)
  + status                = "Active"
  + user                  = "Charles"
}
```

```
# aws_iam_access_key.users_access_keys["Clark"] will be created
+ resource "aws_iam_access_key" "users_access_keys" {
  + create_date           = (known after apply)
  + encrypted_secret       = (known after apply)
  + encrypted_ses_smtp_password_v4 = (known after apply)
  + id                    = (known after apply)
  + key_fingerprint       = (known after apply)
  + secret                = (sensitive value)
  + ses_smtp_password_v4  = (sensitive value)
  + status                = "Active"
}
```

- task7_terraform_output.png

```
@23-22411-057-sudo → ~/Lab13 $ terraform output
all_access_key_secrets = <sensitive>
all_users_details = {
  "Andy" = {
    "access_key_id" = "AKIAVHLXGUTLGF5AYK6"
    "user_arn" = "arn:aws:iam::359416636630:user/users/Andy"
    "user_unique_id" = "AIDAVHLXGUTLP7C7DQ7KK"
  }
  "Angela" = {
    "access_key_id" = "AKIAVHLXGUTLFYLTXNNQ"
    "user_arn" = "arn:aws:iam::359416636630:user/users/Angela"
    "user_unique_id" = "AIDAVHLXGUTLGGXZGPTME"
  }
  "Charles" = {
    "access_key_id" = "AKIAVHLXGUTLIN27S57Z"
    "user_arn" = "arn:aws:iam::359416636630:user/users/Charles"
    "user_unique_id" = "AIDAVHLXGUTLL7MBPW07V"
  }
  "Clark" = {
    "access_key_id" = "AKIAVHLXGUTLOWR4A34K"
    "user_arn" = "arn:aws:iam::359416636630:user/users/Clark"
    "user_unique_id" = "AIDAVHLXGUTLAKFPWJLQ"
  }
  "Creed" = {
    "access_key_id" = "AKIAVHLXGUTLMCXZWVEI"
    "user_arn" = "arn:aws:iam::359416636630:user/users/Creed"
    "user_unique_id" = "AIDAVHLXGUTLOPNVBP4GC"
  }
  "Darryl" = {
    "access_key_id" = "AKIAVHLXGUTLJ3C3PWUN"
    "user_arn" = "arn:aws:iam::359416636630:user/users/Darryl"
    "user_unique_id" = "AIDAVHLXGUTLFDMC SJFQU"
  }
  "David" = {
    "access_key_id" = "AKIAVHLXGUTLJQLABD56"
    "user_arn" = "arn:aws:iam::359416636630:user/users/David"
    "user_unique_id" = "AIDAVHLXGUTLL6GCBCMEG"
  }
  "Dwight" = {
    "access_key_id" = "AKIAVHLXGUTLCSBV2JLI"
    "user_arn" = "arn:aws:iam::359416636630:user/users/Dwight"
    "user_unique_id" = "AIDAVHLXGUTLL3COMCNVU"
  }
  "Erin" = {
    "access_key_id" = "AKIAVHLXGUTLD4SROYGM"
  }
}
```

```

    "user_arn" = "arn:aws:iam::359416636630:user/users/Michael"
    "user_unique_id" = "AIDAVHLXGUTLIJ27H4XKB"
  }
  "Oscar" = {
    "access_key_id" = "AKIAVHLXGUTLIS4LHXUG"
    "user_arn" = "arn:aws:iam::359416636630:user/users/Oscar"
    "user_unique_id" = "AIDAVHLXGUTLPEE7V75UK"
  }
  "Pam" = {
    "access_key_id" = "AKIAVHLXGUTLNR5POAX2"
    "user_arn" = "arn:aws:iam::359416636630:user/users/Pam"
    "user_unique_id" = "AIDAVHLXGUTLJ5BDFTPSO"
  }
  "Peter" = {
    "access_key_id" = "AKIAVHLXGUTLKVU47PXZ"
    "user_arn" = "arn:aws:iam::359416636630:user/users/Peter"
    "user_unique_id" = "AIDAVHLXGUTLMZ3C7PYWW"
  }
  "Phyllis" = {
    "access_key_id" = "AKIAVHLXGUTLHOFDJB3J"
    "user_arn" = "arn:aws:iam::359416636630:user/users/Phyllis"
    "user_unique_id" = "AIDAVHLXGUTLHQMTJJCCT"
  }
  "Robert" = {
    "access_key_id" = "AKIAVHLXGUTLNQHJ2FB"
    "user_arn" = "arn:aws:iam::359416636630:user/users/Robert"
    "user_unique_id" = "AIDAVHLXGUTLHWVYM3CT6"
  }
  "Ryan" = {
    "access_key_id" = "AKIAVHLXGUTLHNNYHS37"
    "user_arn" = "arn:aws:iam::359416636630:user/users/Ryan"
    "user_unique_id" = "AIDAVHLXGUTLG3LOJE4T6"
  }
  "Stanley" = {
    "access_key_id" = "AKIAVHLXGUTLK45W77DZ"
    "user_arn" = "arn:aws:iam::359416636630:user/users/Stamley"
    "user_unique_id" = "AIDAVHLXGUTLLI2XS2AH4"
  }
  "Toby" = {
    "access_key_id" = "AKIAVHLXGUTLEP6U5F6Z"
    "user_arn" = "arn:aws:iam::359416636630:user/users/Toby"
    "user_unique_id" = "AIDAVHLXGUTLPUOUATZFK"
  }
}
@23-22411-057-sudo → ~/Lab13 $ |

```

- task7_tfstate_secrets. Png

```
023-22411-057-sudo → ~/Lab13 $ cat local.tfstate | jq '.resources[] | select(.type=="aws_iam_access_key") | .instances[] | {user: .attributes.user, secret: .attributes.secret}'
{
  "user": "Andy",
  "secret": "7MfrUDx1XfWS/DS+PltpiaCxrHUHmng7K81oWXi3"
}
{
  "user": "Angela",
  "secret": "A9qgxQDgyiYeCLjKe4H6EiWn9XBR/1LESF6YZnL"
}
{
  "user": "Charles",
  "secret": "krku0mwlgseotJAuTGlgTIFh8iYboyvE0XNpIiTwU"
}
{
  "user": "Clark",
  "secret": "xWGxjT0tj6ezZRCy3tXFA69QlXwTY1HEmHzX919t"
}
{
  "user": "Creed",
  "secret": "a3lXINc2Wvy3jbfJvz3XPIDzQyfvVUvPmxE0zYvi"
}
{
  "user": "Darryl",
  "secret": "ziP8sFXVHbea74Q89rh5mQTFBDwCitkXW24L8btJ"
}
{
  "user": "David",
  "secret": "oWLfUSAQXZgQ9XmS18nKaA1kgeidU5uq0Rw1Ak5k"
}
{
  "user": "Dwight",
  "secret": "GJRAttpG5QKCZe2QQg19k5Pz2nBizuWC9FohbCX/"
}
{
  "user": "Erin",
  "secret": "9rj*mwT96d8okauTX1V00TmM/CsvoffFUL7C+V8E8"
}
{
  "user": "Gabe",
  "secret": "VXfT6b07tsDcGtQgG9LxIgNI6VdCopXC1/Jmya19"
}
{
  "user": "Holly",
  "secret": "nSfqmuzelc/2vhjTVpuvkC0SBW5dc020r5T3Ddz"
}
{
  "user": "Jan",
  "secret": "CdIYlci3+7RS15L+xcpocaGDwcqec0QHQL4SqBhg"
}
{
  "user": "Jim",
  "secret": "BSLWd+0YLX5BIuEQmnJMW+7tXAbS/TH/mvQjpnNo"
}
{
  "user": "Jo",
  "secret": "+kh96Fjz5fkW4GCQ6KmmGhzW2CRR5CbxFITsso2m"
}
{
  "user": "Kelly",
  "secret": "4+v8eiw/4P+9R0nxjXzgmHFFcKSfULKG6Ee2mzA"
}
{
  "user": "Kevin",
  "secret": "zbn0j0NYhiReHq/E1czkkaagYqiSEA34JCAu4Ino"
}
{
  "user": "Meredith",
  "secret": "io3XPUZ+F/n3406NEr3rYSHbgPQv4yul4tWR1EGM"
}
```

```

{
  "user": "Jim",
  "secret": "BSLWd+0YLX5BIuEQmnJKW+7tXAbS/TH/mvQjpnNo"
}
{
  "user": "Jo",
  "secret": "+kh96Fjz5fkM4GCQ6KKmGhzw2CRRSCbxFITsso2m"
}
{
  "user": "Kelly",
  "secret": "4+v8eiw/4P+9R0nxjXzgmHFcKSfFULKG6Ee2mzA"
}
{
  "user": "Kevin",
  "secret": "zbm0j0NYhiReHq/E1czkkaagYqiSEA34JCAu4Ino"
}
{
  "user": "Meredith",
  "secret": "io3XPUZ+F/n3406NEr3rYSHbgPQv4yul4tWRiEGM"
}
{
  "user": "Michael",
  "secret": "NbE8u8IybV6koC6Qfll98763DAIIU4JS3I4KBVs"
}
{
  "user": "Oscar",
  "secret": "knpWZX6oj3GMIdbr0d8YUkwIWBQI7xCd3NuSy9h"
}
{
  "user": "Pam",
  "secret": "o7re4bPJRHGM0uIF6SG99cymK8AN8dD6HrYET0ao"
}
{
  "user": "Peter",
  "secret": "23AdvaGLfOdT6C860k3yesjHD4np3otagXkzRQHg"
}
{
  "user": "Phyllis",
  "secret": "6TtwCwL60jG2l8Zidv4lrYNoXZdSTA+M7sZiSSob"
}
{
  "user": "Robert",
  "secret": "DNQ07YIPXGy5gMmhFMEcd+OGbcHrE+0BuiRpHRvd"
}
{
  "user": "Ryan",
  "secret": "Qk3m37nAtvhFIezBIc45zJVEsS1+4ewbYyEA/wci"
}
{
  "user": "Stanley",
  "secret": "oQXkoG10fylhrYCzt6FhfQn4WnfQkBVd5zJd81kr"
}
{
  "user": "Toby",
  "secret": "TdHYF/RGPEHCqUsq7VRVF5N77AvdwVcPgqvMLJ2D"
}
}
@23-22411-057-sudo → ~/Lab13 $

```

- task7_aws_console_all_users. Png

Policy policy2 updated.

Users (30) Info

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

☐	User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID	Active key age	Actions
☐	Admin	/	0	2 hours ago	-	-	-	Active - AKIAVHLXGUT...	2 hours	Refresh
☐	Andy	/users/	1	-	-	-	-	Active - AKIAVHLXGUT...	12 minutes	Refresh
☐	Angela	/users/	1	-	-	-	-	Active - AKIAVHLXGUT...	12 minutes	Refresh
☐	Charles	/users/	1	-	-	-	-	Active - AKIAVHLXGUT...	12 minutes	Refresh
☐	Clark	/users/	1	-	-	-	-	Active - AKIAVHLXGUT...	12 minutes	Refresh
☐	Creed	/users/	1	-	-	-	-	Active - AKIAVHLXGUT...	12 minutes	Refresh
☐	Darryl	/users/	1	-	-	-	-	Active - AKIAVHLXGUT...	12 minutes	Refresh
☐	David	/users/	1	-	-	-	-	Active - AKIAVHLXGUT...	12 minutes	Refresh
☐	Dwight	/users/	1	-	-	-	-	Active - AKIAVHLXGUT...	12 minutes	Refresh
☐	Erin	/users/	1	-	-	-	-	Active - AKIAVHLXGUT...	12 minutes	Refresh
☐	Gabe	/users/	1	-	-	-	-	Active - AKIAVHLXGUT...	12 minutes	Refresh
☐	Holly	/users/	1	-	-	-	-	Active - AKIAVHLXGUT...	12 minutes	Refresh
☐	Jan	/users/	1	-	-	-	-	Active - AKIAVHLXGUT...	12 minutes	Refresh

- task7_aws_console_group_members. Png

developers Info

Summary

User group name: developers

Creation time: January 06, 2026, 17:46 (UTC+05:00)

ARN: arn:aws:iam::359416636630:group/groups/developers

Users (26) | Permissions | Access Advisor

Users in this group (26)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Search

☐	User name	Groups
☐	Andy	
☐	Angela	
☐	Charles	
☐	Clark	
☐	Creed	
☐	Darryl	
☐	David	
☐	Dwight	
☐	Erin	

- task7_aws_console_user_access_key.png

task7_s3_tfstate_multiple_users.png

The screenshot shows the AWS IAM console for a user named Michael. The left sidebar contains the 'Identity and Access Management (IAM)' menu with options like Dashboard, Access management, Users, Roles, Policies, Identity providers, Account settings, Root access management, Temporary delegation requests, Access reports, Access Analyzer, Resource analysis, Unused access, Analyzer settings, Credential report, Organization activity, Service control policies, and Resource control policies. The main content area shows the user's details under the 'Security credentials' tab. It includes a summary of the user's ARN, console access status, and last console sign-in. Below this, there are sections for 'Console sign-in' (with a link to the console and a button to enable console access), 'Multi-factor authentication (MFA) (0)' (with a button to assign an MFA device), and 'Access keys (1)' (with a button to create an access key).

task7_s3_tfstate_multiple_users.png

The screenshot shows the AWS S3 console for a bucket named 'terraform.tfstate'. The left sidebar contains the 'Amazon S3' menu with options like Buckets, Access management and security, and Storage management and insights. The main content area shows the bucket's details under the 'Properties' tab. It includes an 'Object overview' section with fields for Owner, S3 URI, Amazon Resource Name (ARN), Entity tag (Etag), Object URL, Size, Type, and Key. Below this, there is an 'Object management overview' section with a 'Bucket properties' table and a 'Management configurations' table.

Cleanup

- cleanup_destroy_complete.png

```

023-22411-057-sudo → ~/Lab13 $ terraform destroy -auto-approve
aws_iam_group.developers: Refreshing state... [id=developers]
aws_iam_user.users["Robert"]: Refreshing state... [id=Robert]
aws_iam_user.users["Peter"]: Refreshing state... [id=Peter]
aws_iam_user.users["Creed"]: Refreshing state... [id=Creed]
aws_iam_user.users["Erin"]: Refreshing state... [id=Erin]
aws_iam_user.users["Meredith"]: Refreshing state... [id=Meredith]
aws_iam_user.users["Andy"]: Refreshing state... [id=Andy]
aws_iam_user.users["Oscar"]: Refreshing state... [id=Oscar]
aws_iam_user.users["Jim"]: Refreshing state... [id=Jim]
aws_iam_user.users["Kevin"]: Refreshing state... [id=Kevin]
aws_iam_user.users["Jo"]: Refreshing state... [id=Jo]
aws_iam_user.users["Jan"]: Refreshing state... [id=Jan]
aws_iam_user.users["Michael"]: Refreshing state... [id=Michael]
aws_iam_user.users["Ryan"]: Refreshing state... [id=Ryan]
aws_iam_user.users["Stanley"]: Refreshing state... [id=Stanley]
aws_iam_user.users["David"]: Refreshing state... [id=David]
aws_iam_user.users["Holly"]: Refreshing state... [id=Holly]
aws_iam_user.users["Pam"]: Refreshing state... [id=Pam]
aws_iam_user.users["Dwight"]: Refreshing state... [id=Dwight]
aws_iam_user.users["Darryl"]: Refreshing state... [id=Darryl]
aws_iam_user.users["Kelly"]: Refreshing state... [id=Kelly]
aws_iam_user.users["Gabe"]: Refreshing state... [id=Gabe]
aws_iam_user.users["Toby"]: Refreshing state... [id=Toby]
aws_iam_user.users["Phyllis"]: Refreshing state... [id=Phyllis]
aws_iam_user.users["Angela"]: Refreshing state... [id=Angela]
aws_iam_user.users["Clark"]: Refreshing state... [id=Clark]
aws_iam_user.users["Charles"]: Refreshing state... [id=Charles]
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Refreshing state... [id=developers-20260106124648863600000002]
aws_iam_group_policy_attachment.change_password: Refreshing state... [id=developers-20260106124648824000000001]
aws_iam_access_key.users_access_keys["Jo"]: Refreshing state... [id=AKIAVHLXGUTLFAEQ5AGM]
aws_iam_access_key.users_access_keys["Holly"]: Refreshing state... [id=AKIAVHLXGUTLKCMMHGHP]
aws_iam_access_key.users_access_keys["Oscar"]: Refreshing state... [id=AKIAVHLXGUTLIS4LHXUG]
aws_iam_access_key.users_access_keys["Gabe"]: Refreshing state... [id=AKIAVHLXGUTLXXAKHFVT]
aws_iam_access_key.users_access_keys["Dwight"]: Refreshing state... [id=AKIAVHLXGUTLCSBV2JLI]
aws_iam_access_key.users_access_keys["Andy"]: Refreshing state... [id=AKIAVHLXGUTLGFW5AYK6]
aws_iam_access_key.users_access_keys["Clark"]: Refreshing state... [id=AKIAVHLXGUTLOWR4A34K]
aws_iam_access_key.users_access_keys["Ryan"]: Refreshing state... [id=AKIAVHLXGUTLHNNYHS37]
aws_iam_access_key.users_access_keys["Kevin"]: Refreshing state... [id=AKIAVHLXGUTLGBNSQ065]
aws_iam_access_key.users_access_keys["Kelly"]: Refreshing state... [id=AKIAVHLXGUTLETY6JMZU]
aws_iam_access_key.users_access_keys["Phyllis"]: Refreshing state... [id=AKIAVHLXGUTLHOFDJB3J]

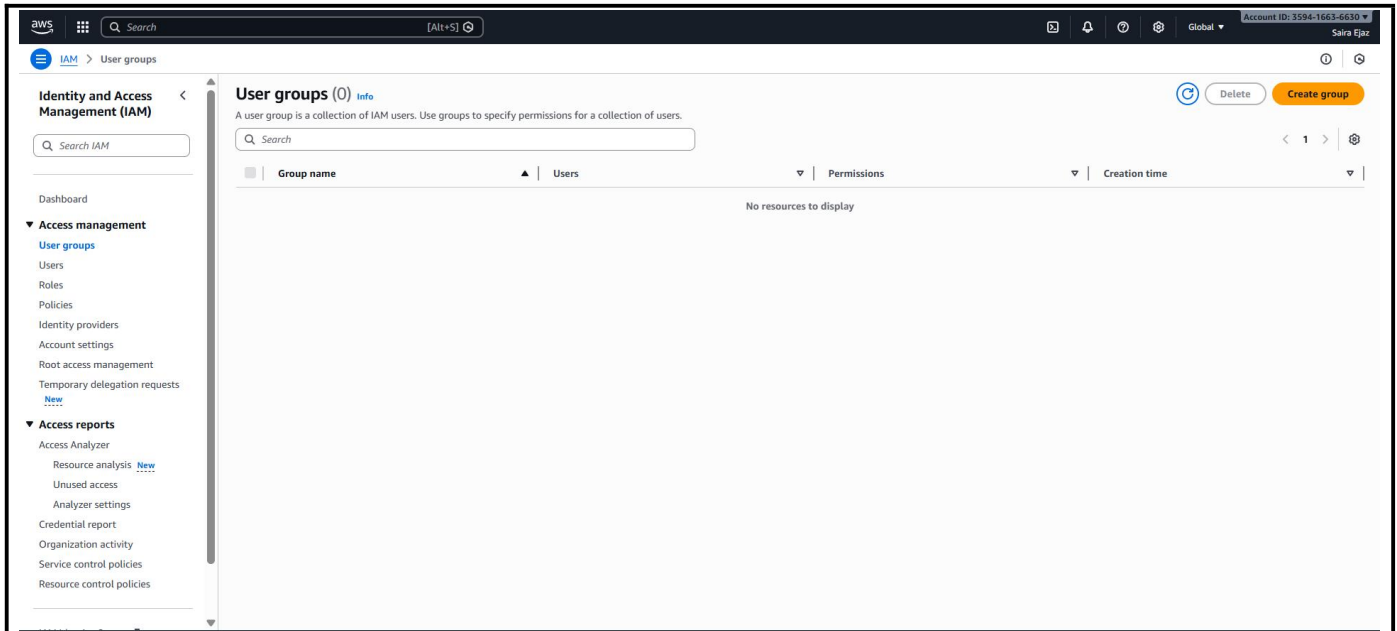
```

- cleanup_aws_console_users_deleted.png

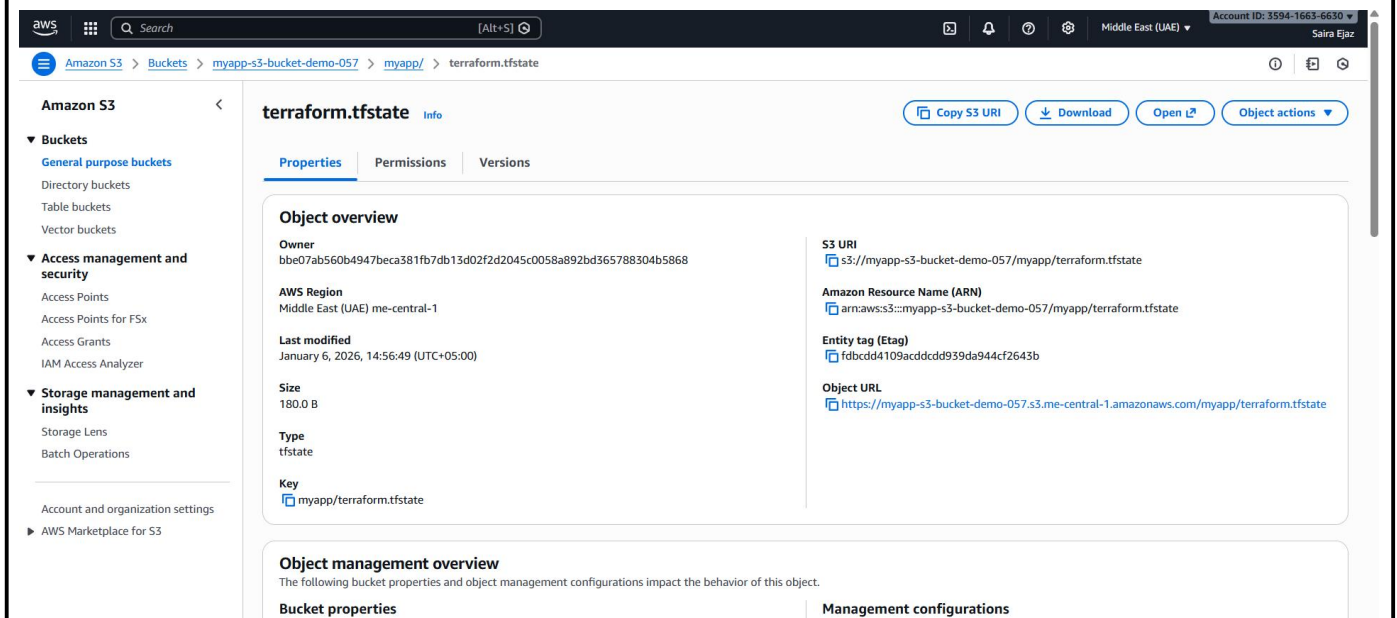
The screenshot shows the AWS IAM console interface. On the left is a navigation sidebar with options like Dashboard, Access management, Access reports, and various IAM tools. The main content area is titled 'Users (4)' and includes a search bar and a table of users. The table has columns for User name, Path, Group, Last activity, MFA, Password age, Console last sign-in, Access key ID, Active key age, and Access status. The users listed are Admin, terraform-admin, User2, and User3, all of which are active and have access keys.

User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID	Active key age	Access status
Admin	/	0	2 hours ago	-	-	-	Active - AKIAVHLXGUT...	2 hours	2
terraform-admin	/	0	3 minutes ago	-	-	-	Active - AKIAVHLXGUT...	3 hours	3
User2	/	0	3 days ago	-	-	-	Active - AKIAVHLXGUT...	4 days	3
User3	/	0	3 hours ago	-	-	-	Active - AKIAVHLXGUT...	3 days	3

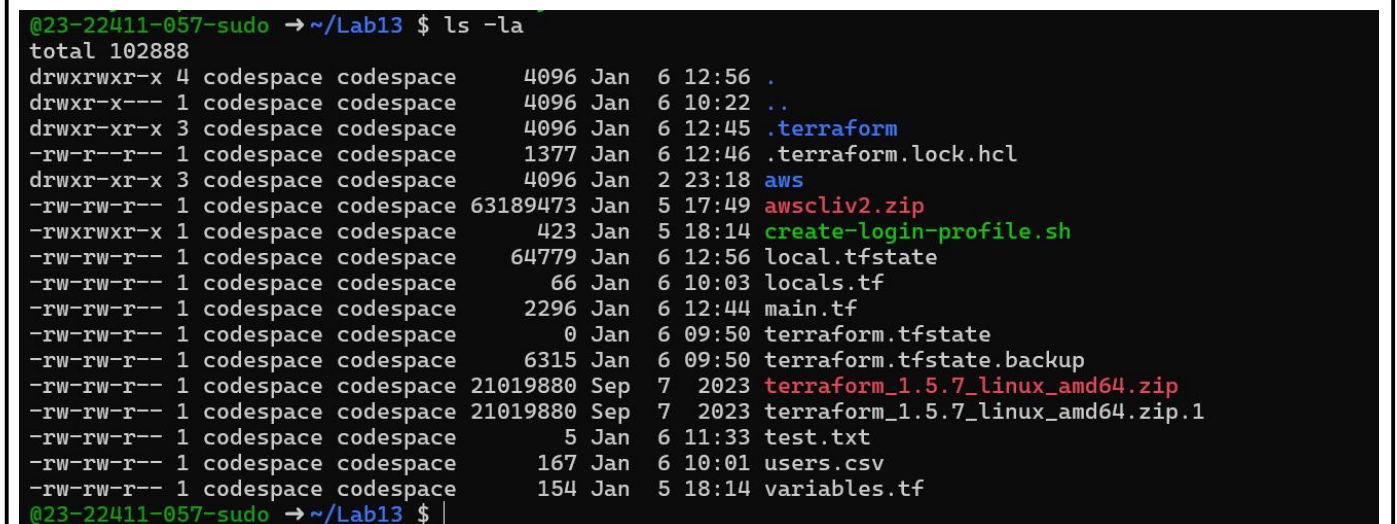
- cleanup_aws_console_group_deleted.png



- cleanup_s3_empty_state.png



- cleanup_final_files.png



- cleanup_s3_bucket_deleted.png (optional)

aws

Search

[Alt+S]

CloudShell

Feedback

Console Mobile App

Middle East (UAE)

Account ID: 3594-1663-6630

Saira Ejaz

Successfully deleted bucket "myapp-s3-bucket-demo-057"

Storage

Amazon S3

Store and retrieve any amount of data from anywhere

Amazon S3 is an object storage service that offers industry-leading scalability, data availability, security, and performance.

Create a bucket

Every object in S3 is stored in a bucket. To upload files and folders to S3, you'll need to create a bucket where the objects will be stored.

Create bucket

Pricing

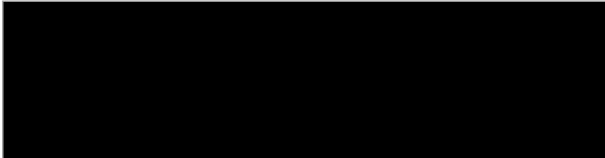
With S3, there are no minimum fees. You only pay for what you use. Prices are based on the location of your S3 bucket.

Estimate your monthly bill using the [AWS Simple Monthly Calculator](#)

[View pricing details](#)

Resources

How it works



CloudShell Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Repository Link

<https://github.com/23-22411-057-sudo/Lab13>